

JYOTHI SWAROOPA JINKA

Hyderabad, India | +91-9390452175

✉ jinka.jyothiswaroopa@gmail.com  [LinkedIn](#)  [GitHub](#)

Education

International Institute of Information Technology, Hyderabad (IIIT-H)

2023 – Present

M.S. by Research in Computer Vision

CGPA: 8.5/10

Chaitanya Bharathi Institute of Technology (CBIT), Hyderabad

2019 – 2023

B.E. in Information Technology

CGPA: 8.54/10

Experience

BharatGen – National AI Mission

Jan 2025 – Present

Generative AI Research Intern

IIIT Hyderabad

- Co-developed **Patram-7B-Instruct**, a multilingual vision-language model for document understanding, integrating OCR, layout parsing, question answering, classification, and summarization across **Indian languages**.
- Built scalable data pipelines for synthetic QA generation, key-value extraction, and summarization, producing **500K+ multimodal training samples** for instruction tuning and supervised fine-tuning.
- Designed distributed training workflows using **DeepSpeed ZeRO-3** on **64 H100 GPUs**.
- Improved training scalability and experiment reproducibility through modular preprocessing, dataset versioning, and configurable training pipelines for large document AI workloads.

Center for Visual Information Technology (CVIT), IIIT Hyderabad

Jan 2024 – Present

Graduate Researcher – Computer Vision & Document AI

- Built ML systems for grounded document question answering, textual attribute recognition, and multimodal document understanding, with a focus on scalable training, evaluation, and inference.
- Developed **TexTAR** and the **MMTAD** dataset for multilingual text attribute recognition in document images; work received the **Best Student Paper Award** at ICDAR 2025.
- Published **3 research papers** in document AI at computer vision venues, spanning document grounding and multilingual tasks.

MathWorks

Jun 2022 – Aug 2022

Engineering Development Group Intern

Hyderabad

- Built **m2doc**, an automated documentation generation tool in MATLAB, reducing manual documentation effort.
- Refactored legacy JavaScript workflows from callbacks/promises to **async/await**, improving code maintainability and developer productivity.
- Worked on internal tooling and automation with emphasis on code quality, readability in production codebases.

Publications

TexTAR: Textual Attribute Recognition in Multi-domain & Multi-lingual Document Images

ICDAR 2025

Best Student Paper Award | [Project Page](#)

Proposes a context-aware transformer for word-level textual attribute recognition (bold, italic, etc.) across multilingual and multi-domain documents.

M3Grounder: Mask-Based Multi-Span and Multi-Granular Grounding for Document QA

CVPR 2026

[Project Page](#)

Introduces a mask-based grounding framework that localizes multi-span evidence at phrase, line, and block levels for document question answering

Projects

Scalable Vision-Language Document Inference with vLLM | *Python, PyTorch, vLLM*

- Designed and implemented high-throughput inference pipelines for vision-language document models using **vLLM**, scaling to handle over **2M+ documents**.
- Improved inference efficiency by achieving a **2.5×** latency reduction through optimized prompt batching, reuse of cached image embeddings, and adaptive request scheduling.
- Conducted comprehensive benchmarking of throughput, memory usage, and output quality across multiple configurations for document QA tasks.

Document QA and Key-Value Extraction Pipeline | *Python, PyTorch, OCR, Transformers*

- Designed an end-to-end pipeline for document question answering and key-value extraction using OCR, layout-aware features, and multimodal transformers.
- Integrated synthetic data generation, grounding, and evaluation workflows to support scalable experimentation on document intelligence tasks.

Multi-Page Document Grounding with Agentic RAG | *Python, PyTorch, RAG, Vision-Language Models*

- Developed a multi-page document QA framework by integrating **M3Grounder** with a retrieval-driven pipeline to handle long-form documents with cross-page evidence.
- Implemented a retrieval module to surface relevant pages and a grounding-centric generation module that links answers to fine-grained visual-textual regions.
- Formulated a unified pipeline combining retrieval and grounded generation to improve answer reliability and provide interpretable evidence across pages.

Signature Verification System | *Python, PyTorch, Computer Vision*

- Built a Siamese Network for signature verification achieving **95%** accuracy, with data augmentation and a web interface.

Context-Aware Music Recommendation with Session Modeling | *Python, Scikit-learn, SQL, Flask*

- Built a hybrid retrieval system blending collaborative signals and audio feature embeddings on the **Million Song Dataset**.

RefChecker – Hallucination Detection for Document QA | *Python, PyTorch, NLP*

- Developed a hallucination detection framework for document-grounded LLM responses, achieving **90%** precision on curated datasets.
- Implemented document-aware verification, modular scoring, and human-in-the-loop evaluation tools.

Technical Skills

Programming: Python, C++, C, SQL, JavaScript

Deep Learning: PyTorch, Hugging Face Transformers, DeepSpeed, vLLM

Systems: Docker, Git, Linux, Weights & Biases, MongoDB

Core Areas: Computer Vision, Multimodal Learning, Distributed Training, Model Optimization,